Multimodal Large Language Models for Video Information Extraction: Survey and Future Directions

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Outline

A Survey on Video Temporal Grounding with Multimodal Large Language Model

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Introduction

Abstract

With the recent advancements in Multimodal Large Language Models (MLLMs), and proliferation of **untrimmed video content** on the Internet, `Video Temporal Grounding` (VTG) has emerged as a critical task in the field of **video understanding** and **information extraction**. In this paper, they proposed a systematical survey of the current research on `VTG-MLLMs` through **three-dimensional taxonomy**:

- **Functional Roles of MLLMs**: They highlighted their architectural significance
- **Training Paradigms**: They analyzed strategies for **temporal reasoning** and **task adoptation**
- **Video Feature Processing**: They discussed **spatiotemporal representation effectiveness**.

Introduction

Overview

There are many applications of VTG-MLLMs:

- Moment Retrieval
- Scene Editing
- Video Question Answering
- **Temporal Question Answering**

Our demand is not only what events occur but also when they **precisely** happen. And to address this gap, Video Temporal Grounding (VTG) has emerged as a pivotal research area. And it involves **localizing video segments** that correspond to **natural language questions**, enabling detailed interaction with video content.

Introduction

Core Challenge

The core challenge of `VTG` lies in precisely **aligning complex linguistic semantics** with `temporal distributed visual information`, while simultaneously handling complex `temporal relationships` within the video.

Tasks

`VTG` encompasses several distinct tasks:

- **Video Moment Retrieval (VMR)**: To identify video segments matching NL descriptions.
- **Dense Video Captioning (DVC)**: To generate temporally aligned captions for multiple events.
- **Video Highlight Detection (VHD)**: To select segments most relevant to the query.
- **Temporally Grounded Video Question Answering (TG-VQA)**: To pinpoint the temporal evidence.

Overview Timeline of Representative VTG-MLLMs

Pre-2024	Q1-2024	Q2-2024	Q3-2024	Q4-2024		2025-Now
SeViLA *(NeurIPS 2023)*	Momentor *(ICML 2024)*	LMR *(Arxiv)*	EI-VLG *(Arxiv)*	ChatVTG *(CVPRW 2024)*	LLaVA-MR *(Arxiv)*	LLaVA-ST *(Arxiv)*
LLaViLo *(ICCV 2023)*	HawkEye *(Arxiv)*	VTG-LLM *(AAAI 2025)*	GeLM *(AAAI 2025)*	Ground-VideoLLM *(Arxiv)*	ReVisionLLM *(CVPR 2025)*	MeCo *(Arxiv)*
VTimeLLM *(CVPR 2024)*	LITA *(ECCV 2024)*	SlowFocus *(NeurIPS 2024)*	TFVTG *(ECCV 2024)*	TimeSuite *(ICLR 2025)*	Seq2Time *(CVPR 2025)*	Time-R1 *(Arxiv)*
TimeChat *(CVPR 2024)*	TimeCraft *(ECCV 2024)*	Mr.BLIP *(Arxiv)*	DeVi *(Arxiv)*	TRACE *(ICLR 2025)*	NumPro *(CVPR 2025)*	VideoExpert *(Arxiv)*

Fig. 1: An overview timeline of representative VTG-MLLMs, organized by initial arXiv release dates.

Introduction

Motivation

Motivated by the rapid advancements in `MLLMs`, unlike traditional approaches relying solely on **visual backbones** with task-specific heads, `VTG-MLLMs` utilize **general-purpose MLLMs** to reason about *temporal relationships, align semantics* and *localize relevant video segments* either directly or indirectly.

Video Temporal Grounding

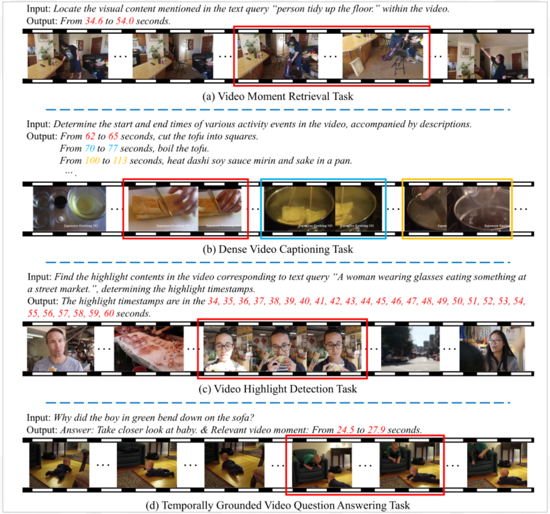
Video Moment Retrieval (VMR)

`VMR` aims to **identify** and **localize** temporal segments within untrimmed videos based on NL queries. It not only requires **accurate alignment** of textual descriptions with specific video segments, but also the **ability to map** video content to precise **temporal boundaries**, testing understanding of fine-grained temporal relationships and semantic conherence.

Dense Video Captioning (DVC)

`DVC` involves generating detailed, temporally grounded descriptions for all significant events or actions occuring in a video, along with their corresponding start and end timestamps. Unlike traditional MR, `DVC` captures the complete narrative by **identifying multiple events** and their temporal dependencies. It also challenges models to manage **overlapping events**, a capability essential for achieving comprehensive fine-grained video understanding.

Video Temporal Grounding



Hình 1 – An illustration of the Video Temporal Grounding (VTG) task..

Video Temporal Grounding

Video Highlight Detection

`VHD` aims to identify **keyframes** or **short segments** within a video that best match a given NL query. Unlike VMR and VDC, it emphasizes **frame-level precision** and evalute the ability of model to accurately pinpoint salient video clips that closely correspond to prompts and to access their contextual significance.

Temporally Grounded Video Question Answering (TG-VQA)

`TG-VQA` requires models to not only provide answer questions bu also identify and localize the precise temporal intervals containing relevant visual evidence. It introduces the added complexity of integrating temporal localization with multimodal reasoning.

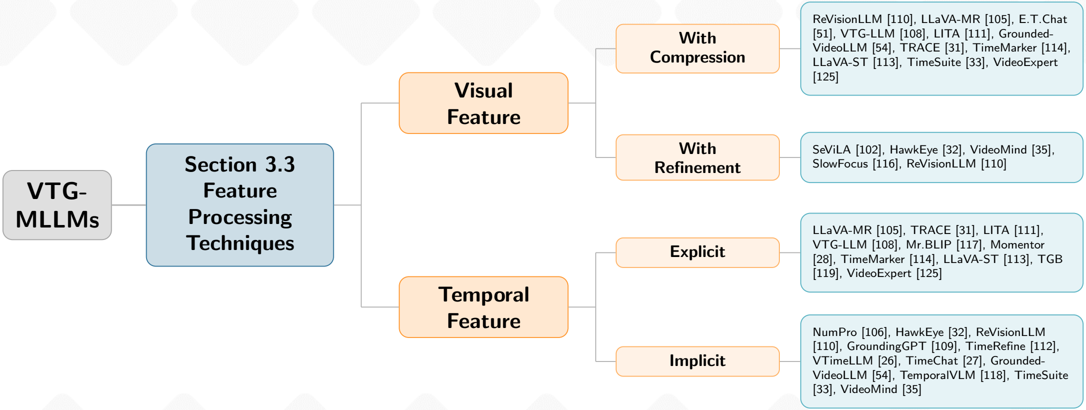
Section 3.1 – Functional Roles of MLLMs

- **VTG-MLLMs**
  - **Section 3.1 Functional Roles of MLLMs**
    - **Facilitator**
      - ChatVTG [29], TEA [91], TimeCraft [56], GroundVQA [91], TFVTG [92], Grounding-Prompter [93], VTG-GPT [94], VERIFIED [95], EI-VLG [96], VideoLights [97], GPTSee [98], LMR [99], ReCorrect [100], DeVi [32], Moment-GPT [101]
    - **Executor**
      - SeViLA [102], LLaViLo [103], TimeChat [27], MLLM-TA [104], LITA [111], TemporalVLM [118], NumPro [106], TRACE [31], VTimeLLM [26], Seq2Time [107], Grounded-VideoLLM [54], TimeSuite [33], VTG-LLM [108], GroundingGPT [109], ReVisionLLM [110], GeLM [115], TimeRefine [112], TimeMarker [114], Momentor [28], SlowFocus [116], LLaVA-ST [113], TPE-VLLM [120], E.T.Chat [51], MeCo [123], VideoMind [35], SpaceVLLM [122], VideoExpert [124], Time-R1 [124], VideoChat-R1 [126], MUSEG [127]

Section 3.2 – Training Paradigms

- **VTG-MLLMs**
  - **Section 3.2 Training Paradigms**
    - **Pretraining**
      - SeViLA [102], VTimeLLM [26], TimeChat [27], VTG-LLM [108], TemporalVLM [118], Seq2Time [107], VideoChat-TPO [121], Grounded-VideoLLM [54], HawkEye [32], TRACE [31], TimeRefine [112], TimeMarker [114], Momentor [28], ReVisionLLM [110], SlowFocus [116], LLaVA-ST [113], MeCo [123], Time-R1 [124], VideoChat-R1 [126], MUSEG [127]
    - **Fine-Tuning**
      - LLaViLo [103], Mr.BLIP [117], LLaVA-MR [105], VideoLights [97], TGB [119], GPTSee [98], EI-VLG [96], LMR [99], TEA [90], ReCorrect [100]
    - **Training-Free**
      - NumPro [106], ChatVTG [29], DeVi [32], TFVTG [92], VTG-GPT [94], Moment-GPT [103], Grounding-Prompter [95]

Section 3.3 – Feature Processing Techniques



Directions

Four Directions

- **Video Temporal Grounding**: to locate specific moments in a video based on a text query.
- **Video Highlighting Detection**: to identify engagin segments within a given video, which requires to assign a saliency score to each video clip and select the highest-scoring.
- **Dense Video Captioning**: Requires both *event localization* and *captioning* within the same framework.
- **Video Grounding Question Answering**: Requires models to understand video content to answer questions and to locate relevant segments as visual evidence.

Through the PRISm: Importance-Aware Scene Graphs for Image Retrieval

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Abstract

Challenge

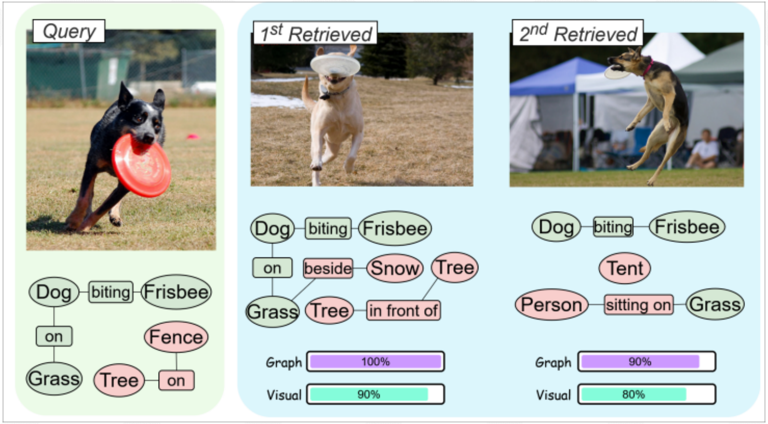
In `Event Retrieval` or `Frame Retrieval` lies at the core of CV tasks. Most of previous frameworks solely rely on `global visual encoders` that do not weight **objects** or **relations**, which overlooked Importance of invidual elements.

Proposed Solution

To address this limitation, they introduce `PRISM` that includes two crucial components.

- **Importance Prediction Module** identifies and retains the most critical objects and relations triplet within an image.
- **Edge-Aware Graph Neural Network** encodes *relational structure* and integrates global visual features to produce semantically informed image embeddings.

Pipeline



Hình 2 – Top-2 image-to-image visual features, pruning graphs to retain the most important objects and relations, and then encodes objects interactions, emphasizing relational structure..

Methodology

Importance Prediction Module

Aims to predict **semantic importance** of both objects and relational triplets `<subject, relation, object>` through visual and textual cues. It also used `Human-generated captions` during training as a proxy for semantic relevance.

Edge-Aware Graph Neural Network

- Refine representation through `GNN` that integrates both semantic and visual information, which fuses multiple modalities into a **unified image representation**.
- Extend standard attention-based GNNs by encoding edge information during message passing.

And it is realized in `Edge-aware Graph Reasoning Module`, which featured in the dual stream. One stream focuses on *Goal Visual Feature*, and other forms *Relational Reasoning*.

Ideas from this paper

Future works

This paper introduced a `image-to-image` retrieval while current direction of team is `text-to-image` retrieval.

- The **edges** should encode `temporal relations` between objects across frames in a video.
- Add a **text encoder stream** on the query side then pair it with a **visual + graph stream**.

TemporalVLM: Video LLMs for Temporal Reasoning in Long Videos

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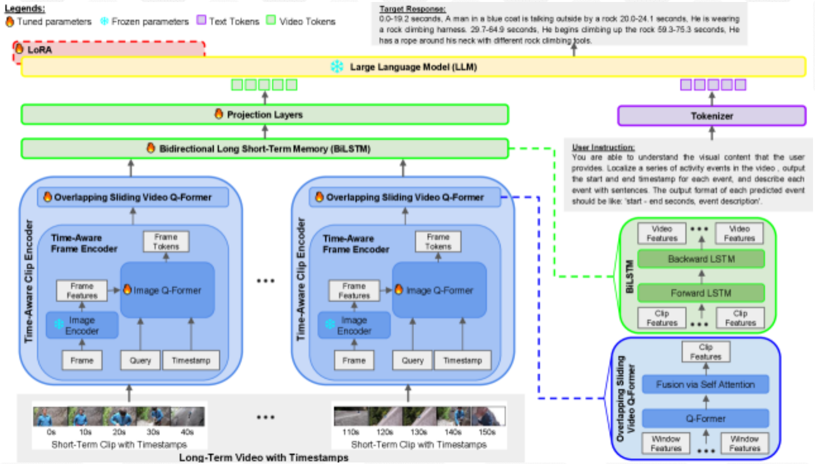
Abstract

Proposed Solution

`TemporalVLM` is a video LLM model for temporal reasoning and fine-grained understanding in long videos. It inclues a `visual encoder` for mapping a long-term video into features which are **time-aware** and contain both **local** and **global** cues.

- Divide video into **short-term clips**, jointly encoded with *timestamps* and **fused** across overlapping temporal windows.
- Local features are passed through **Bidirectional Long Short-term Memory (BiLSTM)** module for global feature aggeration.
- They also introduce a large-scale long video datasets.

Pipeline



Hình 3 – TemporalVLM includes two novel components: a time-aware clip encoder for extracting time-aware fine-grained cues and a BiLSTM module for capturing long-range temporal dependencies..

Contributions

Short-Term Temporal Reasoning

They introduce a **time-aware** clip encoder for extracting fused local fine-grained cues within a clip.

- Divide long-term video into short-term clips. Then **sampled frames** along with timestamps are then passed to **time-aware clip encoder** for jointly encoding frame contents and timestamps.
- They used a pre-trained image encoder to obtain frame features, which are jointly encoded with timestamps via an image `Q-Former`.

Long-Term Temporal Reasoning

Bidirectional Long Short-Term Memory, a module for computing global features across multiple clips.

- Concatenate **time-aware** local features extracted from clips in temporal order.
- Pass the sequence of local features to the BiLSTM for aggerating global features.

VideoExpert: Augmented LLM for Temporal-Sensitive Video Understanding

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Abstract

Challenge

MLLMs tend to rely more on **language patterns** than visual cues when generating timestamps.

Proposed Solution

`VideoExpert`, an MLLM for serveral **temporal-sensitive** video tasks.
It has twoo parallel modules:

- **Temporal Expert** models time sequences and performing temporal grounding. It compresses tokens to capture dynamic variations in videos and predict **event localization**. **Spatial Expert** focuses on content detail analysis. It handles designed spatial tokens and language input to generat content-related responses.

Pipeline

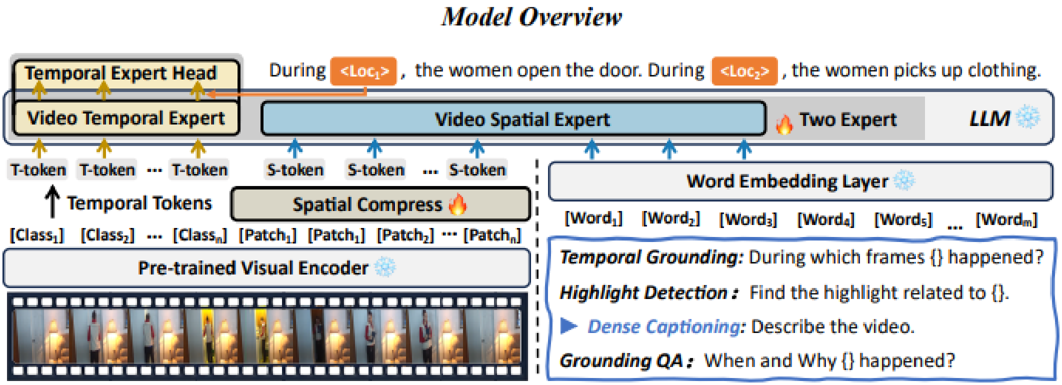

Fig. 3: Overview of the proposed VideoExpert for a series of temporal-sensitive video tasks.

Hình 4 – VideoExpert includes two parallel modules: **Temporal Expert** and **Spatial Expert**..